

CAP Theorem

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Overview

- The **CAP Theorem** states that a distributed system can guarantee at most **two out of the following three properties** simultaneously:
- **C – Consistency**
- **A – Availability**
- **P – Partition Tolerance**
- The theorem was proposed by **Eric Brewer** and later formally proven by **Seth Gilbert** and **Nancy Lynch**.

1. Consistency (C)

- Every read receives the most recent write or an error.
- **Example**
 - Suppose Account Balance = ₹10,000.
 - After a withdrawal: Balance = ₹9,000
- Every server immediately returns ₹9,000.
- No user sees stale data.

2. Availability (A)

- Every request receives a response, even if some nodes have failed.
- **Example**
 - An e-commerce website continues serving users even when some servers are down.
 - Users always get a response.
- Although the information may not be the latest.

3. Partition Tolerance (P)

- The system continues operating despite network failures that prevent communication between nodes.

Example

- Node A $\leftrightarrow$ Node B
- A network partition occurs.
- Even though A and B cannot communicate, the system must continue functioning.

Why cannot we have all three?

- During a network partition, the system must choose:
- **Option 1: Consistency**
 - Wait until replicas agree.
 - Result: Correct data, Some requests rejected
- **Option 2: Availability**
 - Respond immediately.
 - Result: Service remains available, Some responses may contain stale data
- Thus, during a partition, a distributed system must choose between **Consistency** and **Availability**.

CAP Trade-offs

1. CP (Consistency + Partition Tolerance)

• Characteristics

- Data is always consistent.
- System tolerates network partitions.
- Some requests may be denied during failures.

• **Example:** Distributed banking system

- Node A balance = ₹9000

Node B disconnected

• To avoid inconsistency:

- Read request → Rejected/Delayed until synchronization occurs.

Advantages

- ✓ Strong consistency
- ✓ Correct results

Disadvantages

- ✗ Reduced availability during partitions

Examples

- Apache HBase
- MongoDB (with majority write concerns)
- Google Spanner

CAP Trade-offs

contd...

2. AP (Availability + Partition Tolerance)

• Characteristics

- System always responds.
- Tolerates network failures.
- Data may be temporarily inconsistent.

• Example: Social media likes

- Server A = 101 likes
Server B = 100 likes
- A user connected to A sees: 101 likes
- Another user connected to B sees: 100 likes
- Eventually both become: 101 likes

Advantages

- ✓ High availability
- ✓ Excellent scalability

Disadvantages

- ✗ Temporary stale data

Examples

- Apache Cassandra
- Amazon DynamoDB
- Riak

CAP Trade-offs

contd...

3. CA (Consistency + Availability)

- **Characteristics**

- Always consistent.
- Always available.
- Does not tolerate partitions.

- **Example:** Single-node relational database

- Client → Database

- No network partition exists.

- The system can provide:

- ✓ Consistency
- ✓ Availability

- **Limitation**

- If a partition occurs, CA cannot be maintained.

Examples

Traditional centralized databases:

- MySQL
- PostgreSQL
(when deployed on a single server)

Real World Examples

System	CAP Choice	Reason
Banking System	CP	Correctness is critical
ATM Network	CP	Avoid incorrect balances
Social Media Feed	AP	Availability is more important
DNS	AP	Fast responses preferred
Single SQL Database	CA	No partition handling required
Global NoSQL Database	AP	High scalability and availability

Thank You